

Geometry of Particle Methods

Lambert et al. 2022: introduce “smoothed particles” for VI.

Goal: for $\pi(x) \propto e^{V(x)}$, estimate $\hat{\pi}$,

$$\arg \inf_{\hat{\pi} \in \mathcal{G}_k} \text{KL}(\hat{\pi} \mid \pi) \quad \mathcal{G}_k := \left\{ \frac{1}{k} \sum_{i=1}^k \mathcal{N}(\mu_i, \Sigma_i) : \mu \in \mathbb{R}^d, \Sigma \in S_+^d \right\}$$

“Wasserstein over Wasserstein” structure: take $\mathcal{M} = \mathcal{G}_1$

$\implies \mathcal{G}_k$ are the k -particle empirical msrs. in $\mathcal{P}(\mathcal{M})$!

further explored: Petit-Talamon et al. 2025, Bonet et al. 2025

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We can also allow *weighted* particles:

$$\hat{\mu}_t := \sum_{i=1}^n w(t, X_i) \delta_{X^{(i)}} \quad \{X^{(i)}\}_{i=1\dots n}$$

Dynamics:

$$\partial_t w(t, X) = \alpha(t, X, \mu_t) \quad \Longleftrightarrow \quad \partial_t \mu(t, X) = \alpha(t, X, \mu_t) \mu(t, X, \mu_t)$$